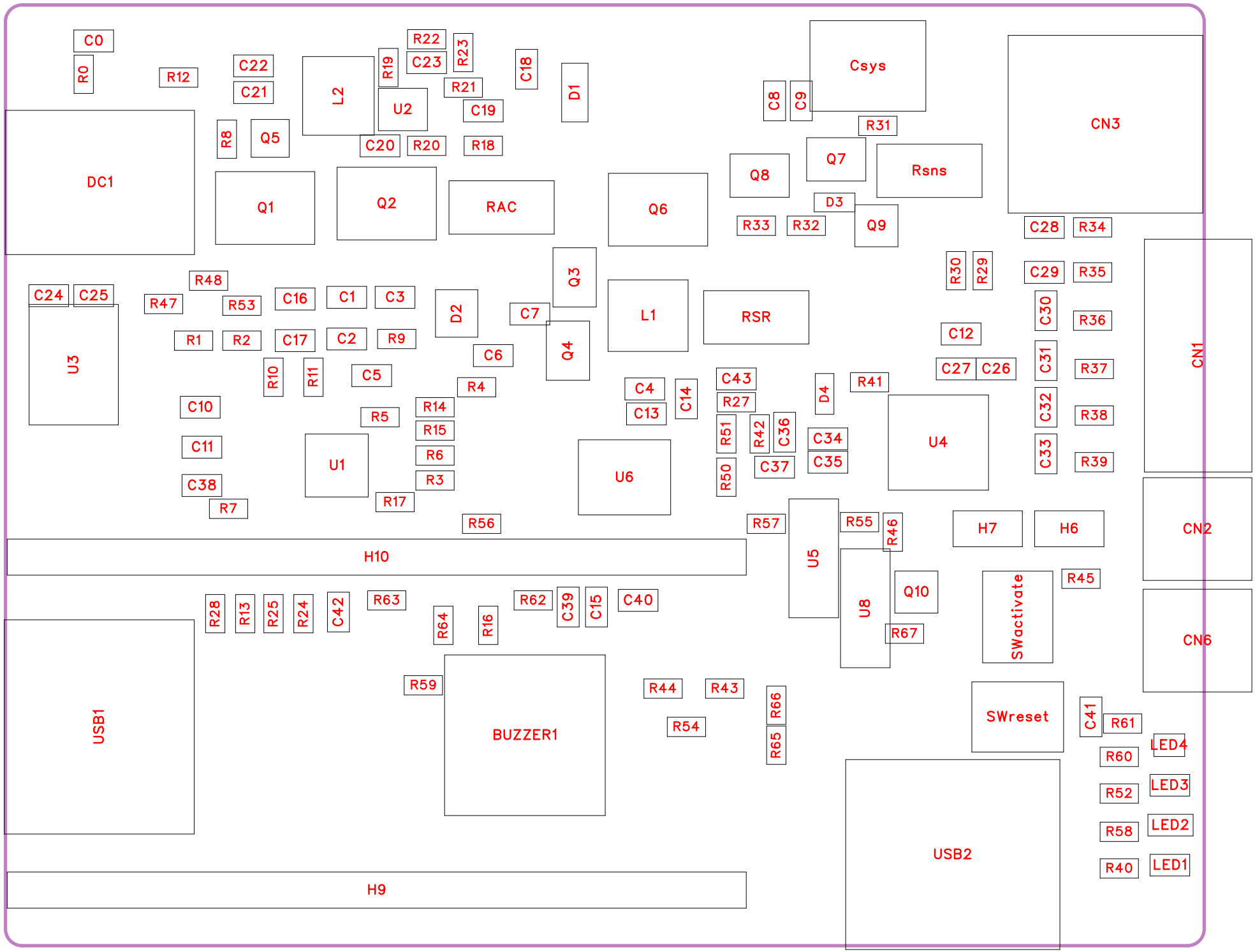
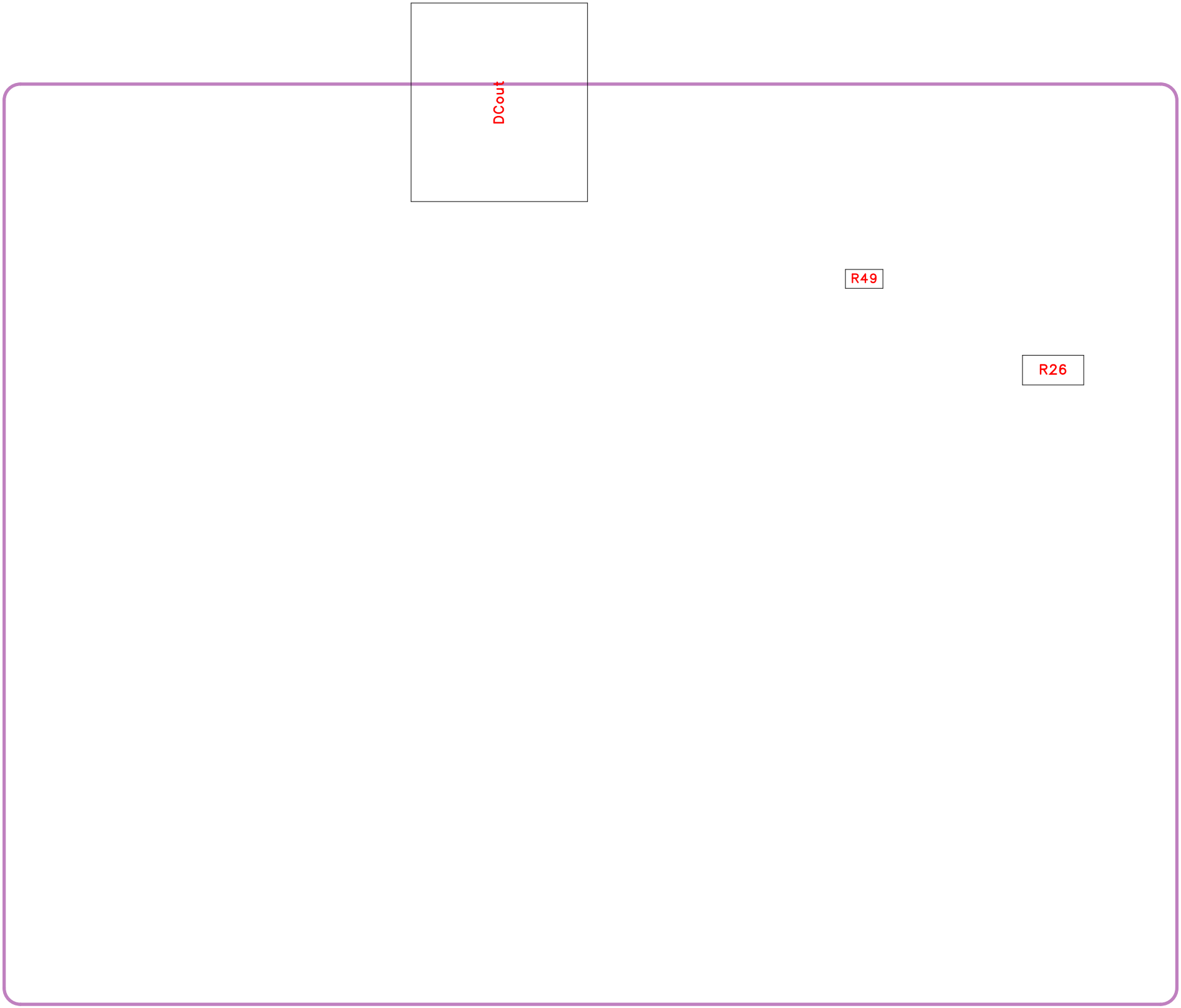

C0
R0
R12
C22
C21
L2
R19
R22
C23
R23
R21
U2
C19
C18
D1
Csys
C8
C9
R31
CN3
DC1
R8
Q5
C20
R20
R18
Q1
Q2
RAC
Q6
Q8
Q7
Rsns
D3
R33
R32
Q9
C28
R34
R30
R29
C29
R35
C24
C25
R48
R47
R53
C16
C1
C3
D2
C7
Q3
L1
RSR
C30
R36
U3
R1
R2
C17
C2
R9
C6
Q4
C12
C31
R37
CN1
R10
R11
C5
R4
C4
C14
C43
D4
R41
C27
C26
C10
R5
R14
C13
R27
C32
R38
R15
R51
R42
C36
C34
U4
C11
U1
R6
U6
C37
C35
C33
R39
C38
R3
R50
R7
R17
R56
R57
R55
R46
H7
H6
CN2
H10
U5
U8
Q10
SWactivate
R45
R28
R13
R25
R24
C42
R63
R64
R16
R62
C39
C15
C40
R67
CN6
R59
BUZZER1
R44
R43
R66
SWreset
C41
R61
USB1
R54
R65
LED4
R60
LED3
R52
LED2
R58
USB2
R40
LED1
H9

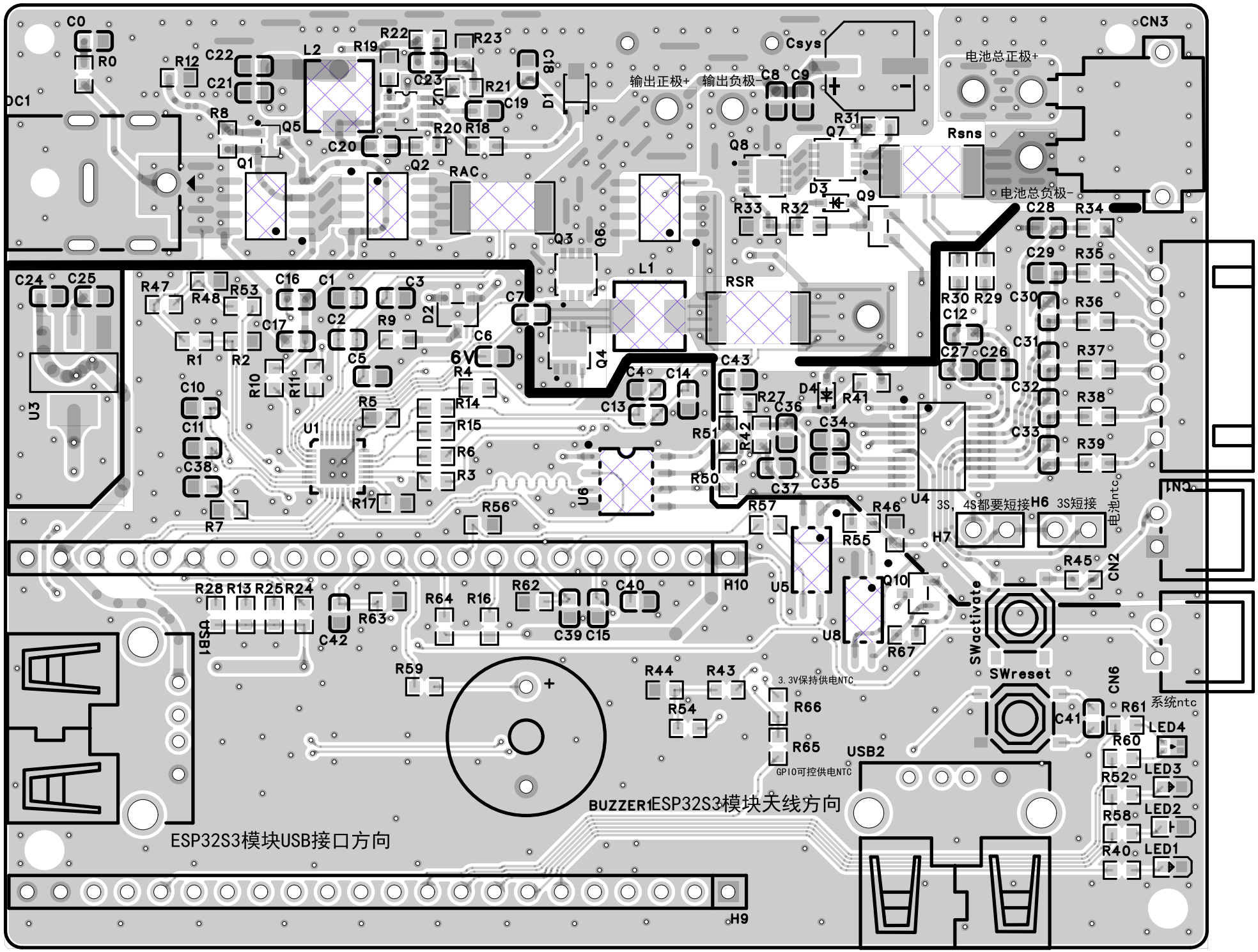

DCout
R49
R26

Quantity	Designator	Footprint	Value	Comment	Device
1	BUZZER1	BUZ-TH_BD12.0-P7.60-D0.6-FD	2.3kHz	2.3kHz	TMB12A03
1	C0	C0603	2.2uF	2.2uF	CL10A225KA8NNNC
20	C1,C2,C3,C4,C12,C13,C14,C18,C19,C20,C26,C27,C28,C29,C30,C31,C32,C33,C41,C43	C0603	100nF	100nF	FCC0603B104K500CT
5	C5,C6,C34,C36,C37	C0603	1uF	1uF	CGA0603X5R105K500JT
2	C7,C16	C0603	47nF	47nF	CL10B473KB8NNNC
4	C8,C9,C24,C25	C0603	10uF	10uF	CGA0603X5R106K100JT,CL10A106MA8NRNC
5	C10,C11,C15,C38,C40	C0603	100pF	100pF	CL10C101JB8NNNC
1	C17	C0603	1nF	1nF	CL10B102KB8NNNC
2	C21,C22	C0603	22uF	22uF	CL10A226MP8NUNE
1	C23	C0603	33pF	33pF	CL10C330JB8NNNC
1	C35	C0603	4.7uF	4.7uF	CGA0603X5R475K250JT
2	C39,C42	C0603	10nF	10nF	CL10B103KB8NNNC
1	CN1	CONN-TH_6P-P2.50_HCZZ0139-6		HX-XH2.54-6PWZ	HX-XH2.54-6PWZ
2	CN2,CN6	CONN-TH_HX-XH2.54-2PWZ		HX-XH2.54-2PWZ	HX-XH2.54-2PWZ
1	CN3	CONN-TH_XT30PW-M		XT30PW-M30.G.Y	XT30PW-M30.G.Y
1	Csys	CAP-SMD_BD6.3-L6.6-W6.6-LS7.3-FD-H7.7	220uF	220uF	JVJ25V220M6x8
1	D1	SOD-123FL_L2.7-W1.8-LS3.8-RD		SMF24A	SMF24A_C19077514
1	D2	SOT-23-3_L2.9-W1.6-P1.90-LS2.8-BR		BAT54C,215	BAT54C,215
2	D3,D4	SOD-323_L1.7-W1.3-LS2.6-RD-1		BAT54WS	BAT54WS_C7502694

1	DC1	DC-IN-TH_HC-DC-007B-2.5		HC-DC-007B-2.5	HC-DC-007B-2.5
1	DCout	CONN-TH_XT30PW-F-1		XT30PW-F20.G.Y	XT30PW-F20.G.Y
2	H6,H7	HDR-TH_2P-P2.54-V-M		PZ254V-11-02P	PZ254V-11-02P
2	H9,H10	HDR-TH_22P-P2.54-V-F		CS-1165SDIG-1*22	CS-1165SDIG-1*22
1	L1	IND-SMD_L5.4-W5.2_FXL05XX	3.3uH	3.3uH	FXL0530-3R3-M
1	L2	IND-SMD_L5.4-W5.2_FXL0530	4.7uH	4.7uH	FXL0530-4R7-M
1	LED1	LED0603-RD		GL0603UG01	GL0603UG01
1	LED2	LED0603-FD		GL0603UR01	GL0603UR01
1	LED3	LED0603-RD_BLUE		GL0603UB01	GL0603UB01
1	LED4	LED-SMD_4P-L1.6-W0.8-RD_1		XL-1608SURUGC	XL-1608SURUGC
3	Q1,Q2,Q6	SOP-8_L5.0-W4.0-P1.27-LS6.0-BL		FDS6680A	FDS6680A
4	Q3,Q4,Q7,Q8	PDFN-8_L3.1-W3.1-P0.65-LS3.4-BL		SIS412DN-T1-GE3-ES	SIS412DN-T1-GE3-ES
1	Q5	SC-70-3_L2.0-W1.3-P0.65-LS2.1-BL		BSS138W	BSS138W_C890266
1	Q9	SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR-1		BSS84	BSS84_C7420340
1	Q10	SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR		AO3401	AO3401_C5224195
1	R0	R0603	2Ω	2Ω	0603WAF200KT5E
2	R1,R18	R0603	499kΩ	499kΩ	0603WAF4993T5E
1	R2	R0603	133kΩ	133kΩ	0603WAF1333T5E
6	R3,R9,R14,R15,R20,R42	R0603	10Ω	10Ω	0603WAF100JT5E
1	R4	R0603	316kΩ	316kΩ	0603WAF3163T5E

3	R5,R44,R48	R0603	100kΩ	100kΩ	0603WAF1003T5E
5	R6,R10,R11,R55,R67	R0603	4.7kΩ	4.7kΩ	0603WAF4701T5E
1	R7	R0603	14.3kΩ	14.3kΩ	FRC0603F1432TS
1	R8	R0603	3MΩ	3MΩ	0603WAF3004T5E
5	R12,R31,R32,R33,R46	R0603	1MΩ	1MΩ	0603WAF1004T5E
12	R13,R16,R24,R25,R28,R41,R45,R50,R51,R62,R63,R64	R0603	10kΩ	10kΩ	FRC0603F1002TS
1	R17	R0603	866kΩ	866kΩ	FRC0603F8663TS
1	R19	R0603	9.09kΩ	9.09kΩ	FRC0603F9091TS
1	R21	R0603	20kΩ	20kΩ	0603WAF2002T5E
2	R22,R56	R0603	40.2kΩ	40.2kΩ	FRC0603F4022TS
1	R23	R0603	7.5kΩ	7.5kΩ	0603WAF7501T5E
1	R26	R1206	0Ω	0Ω	RE1206F1R000
3	R27,R65,R66	R0603	0Ω	0Ω	0603WAF0000T5E
5	R29,R30,R40,R52,R61	R0603	100Ω	100Ω	0603WAF1000T5E
9	R34,R35,R36,R37,R38,R39,R53,R54,R57	R0603	1kΩ	1kΩ	0603WAF1001T5E
2	R43,R47	R0603	910kΩ	910kΩ	FRC0603F9103TS
1	R49	R0603		HNTC0603-103F3450FA	HNTC0603-103F3450FA
2	R58,R60	R0603	330Ω	330Ω	0603WAF3300T5E
1	R59	R0603	47Ω	47Ω	0603WAF470JT5E
2	RAC,RSR	R2512	10mΩ	10mΩ	RE2512F3R010
1	Rsns	R2512	5mΩ	5mΩ	JER2512F3R005
2	SWactivate,SWreset	SW-SMD_4P-L5.1-W5.1-P3.70-LS6.5-TL_H1.5		TS-1187A-B-A-B	TS-1187A-B-A-B

1	U1	WQFN-28_L4.0-W4.0-P0.40-BL-EP		BQ24780SRUYR	BQ24780SRUYR
1	U2	TSOT-23-8_L2.9-W1.6-P0.65-LS2.8-BL		MP2315SGJ-Z	MP2315SGJ-Z
1	U3	SOT-223-3_L6.5-W3.4-P2.30-LS7.0-BR		AMS1117-3.3	AMS1117-3.3_C6186
1	U4	TSSOP-20_L6.5-W4.4-P0.65-LS6.4-BL		BQ7692003PWR	BQ7692003PWR
2	U5,U8	SOP-4_L4.4-W2.8-P1.27-LS7.0-BR		TLP293(GB-TPL,E	TLP293(GB-TPL,E
1	U6	SOIC-8_L4.9-W3.9-P1.27-LS6.0-TL		ISO1640BDR	ISO1640BDR
2	USB1,USB2	USB-A-TH_AF-WJDG		AF 90° WJDG	AF 90° WJDG


C0
R0
DC1
R12
C22
C21
L2
R19
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R23
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U2
R21
C19
C18
D1
R8
Q5
R20
R18
C20
Q1
Q2
RAC
输出正极+
输出负极-
C8
C9
Csys
R31
Q7
Q8
D3
Q9
R33
R32
Rsns
电池总正极+
电池总负极-
CN3
C28
R34
C29
R35
C30
R36
C31
R37
C32
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U4
R7
R17
R56
R57
R46
3S，4S都要短接
H6
3S短接
电池ntc
CN1
H7
R55
R45
CN2
U3
R28
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U8
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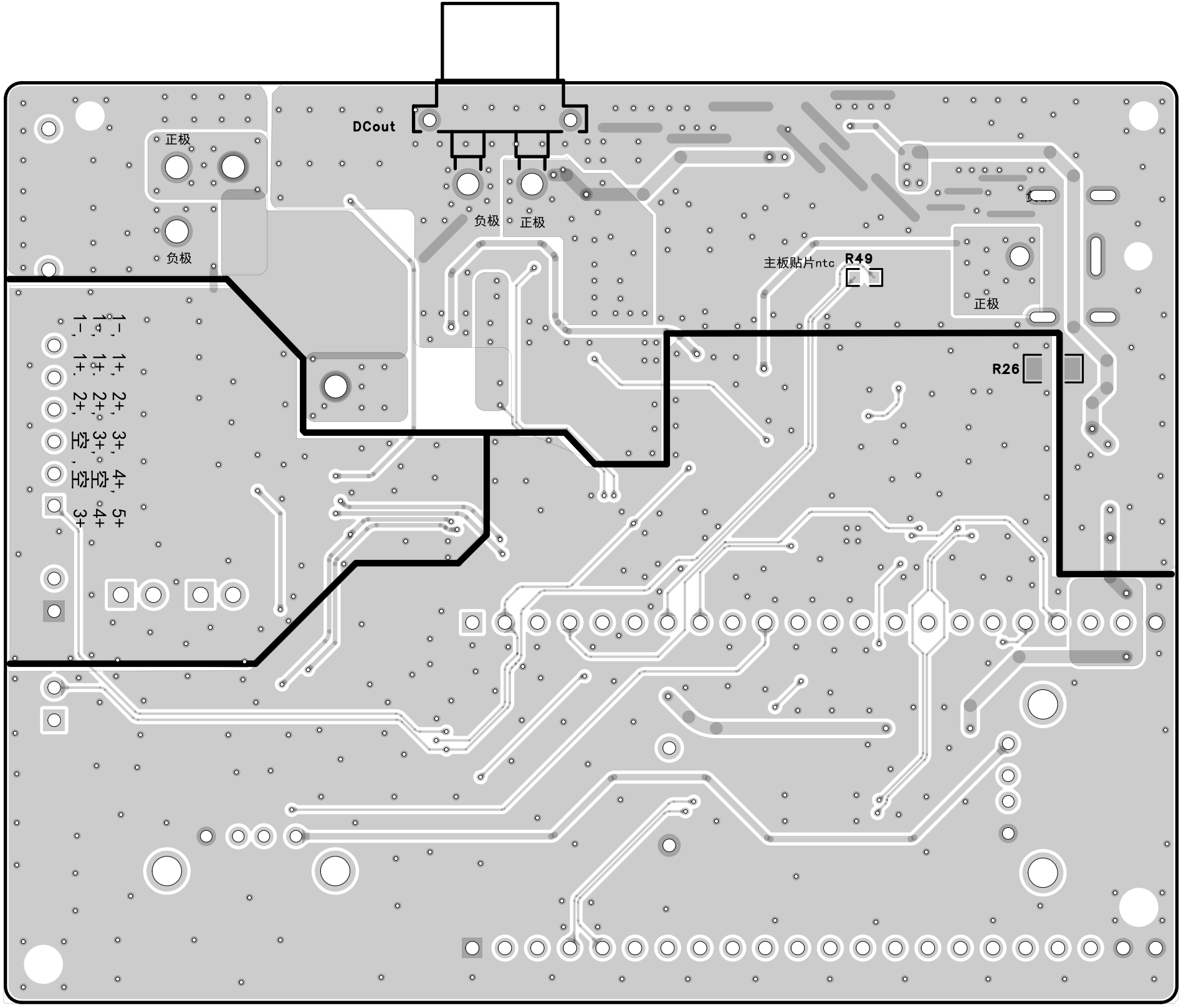
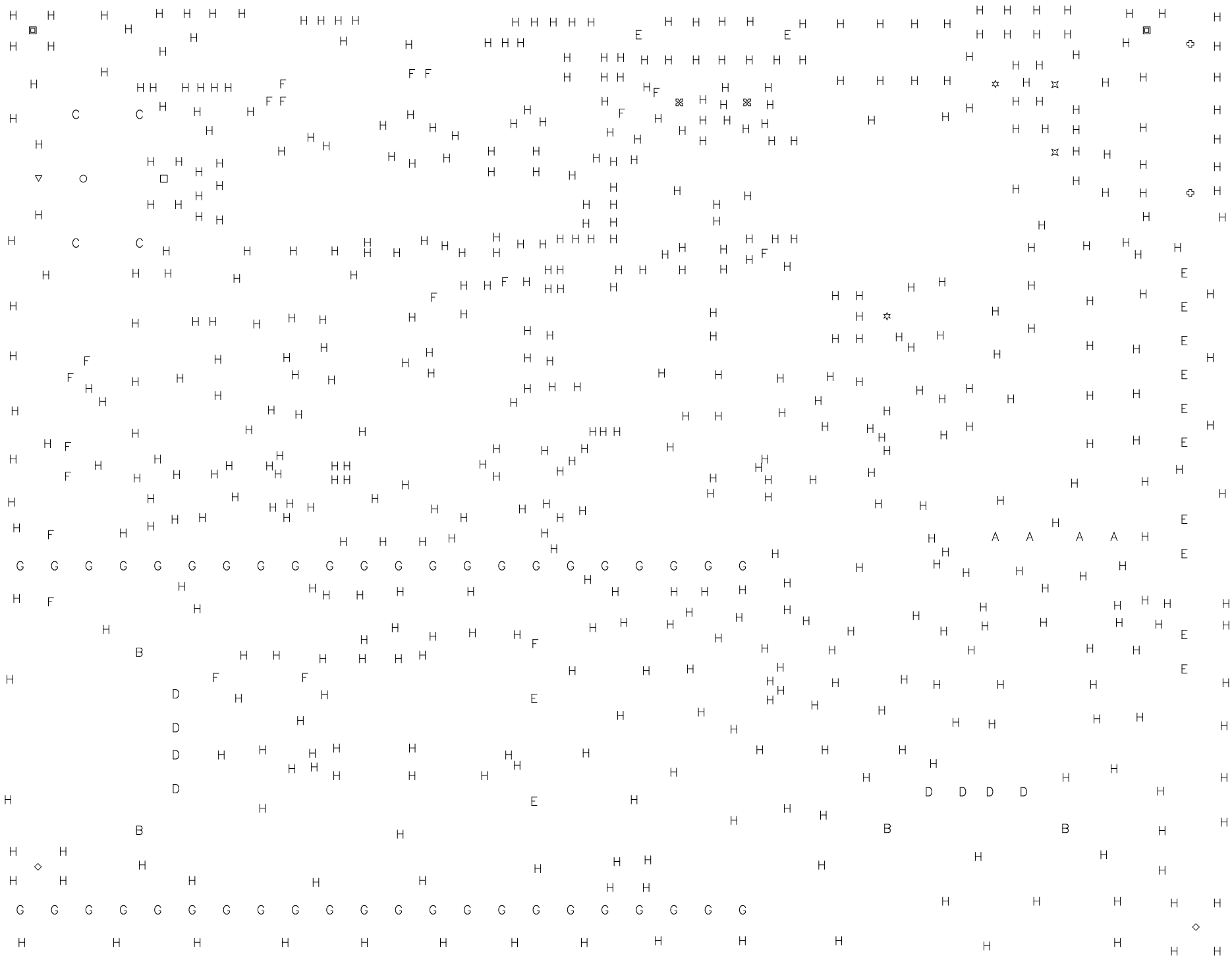
SWactivate
USB1
R59
R44
R43
R54
3.3V保持供电NTC
R66
R65
GPIO可控供电NTC
SWreset
CN6
C41
R61
系统ntc
LED4
R60
LED3
R52
LED2
R58
LED1
R40
USB2
BUZZER1
ESP32S3模块天线方向
ESP32S3模块USB接口方向
H9


DCout
正极
负极
负极
正极
主板贴片ntc
R49
正极
R26
1-, 1+, 2+, 3+, 4+, 5+
1-, 1+, 2+, 空, 空, 4+
1-, 1+, 2+, 3+, 空, 3+